

Recommendation: Major Revision

The manuscript has improved substantially, and I appreciate the considerable effort made in response to the previous reviews. In particular, the revised framing around *trade-aware movement* is much clearer. The manuscript now appropriately distinguishes between PPO-learned movement and the fixed Rule-T exchange mechanism. The observation and reward specifications have also been clarified, and the additional robustness, survival, and inequality analyses materially strengthen the paper. I therefore believe that the revised manuscript now contains a potentially publishable contribution.

I am not yet able to recommend acceptance, however, because the paper’s central identification claim remains incompletely established.

1. The effect of “trade-awareness” is still not cleanly identified.

The principal macroeconomic comparison is between Rule-M movement and PPO movement with MRS information. This comparison can establish that replacing Rule M with the learned movement policy changes economic outcomes. It does not, by itself, establish that these differences are caused specifically by the inclusion of trading-related information (MRS).

The newly added MRS ablation is therefore highly relevant and, in my view, an important improvement. However, it is currently evaluated only through learning diagnostics. The natural completion of this analysis is to evaluate

PPO with MRS versus PPO without MRS

using the same downstream outcomes and matched environment seeds used elsewhere in the paper. At minimum, this should include carrying capacity/survival, trade frequency or volume, price behavior, resource consumption/welfare, and inequality.

This comparison would hold the learning architecture, action space, reward specification, and training procedure fixed while varying the market-information channel itself. It would therefore provide the cleanest evidence for the manuscript’s principal claim that *trade-aware movement*, rather than learned movement more generally, is responsible for the reported effects.

Until such an outcome-level ablation is provided, the paper should either perform this experiment or substantially narrow its causal language.

2. The feasible movement sets appear to differ between the rule-based and PPO conditions.

A second issue concerns the definition of movement itself. Rule M is described as selecting a destination within the agent’s vision range, whereas the PPO action space is given as

$$\mathcal{A} = \{\text{North, South, East, West, Stay}\},$$

with movement of one cell per step.

If this description accurately reflects the implementation, then the two regimes differ not only in the decision rule used to select movement, but also in the feasible movement distance. This distinction could directly affect survival, resource access, spatial concentration, encounter frequency, and consequently trade volume. The authors should clarify precisely how movement distance is implemented in the two conditions. If the feasible movement sets are indeed different, the comparison should be redesigned so that both policies operate over the same feasible destinations. Otherwise, the statement that the two conditions “differ only in the movement policy” is too strong.

3. The Markov/Dec-POMDP discussion still requires technical clarification.

The revision is substantially better in acknowledging partial observability and multi-agent non-stationarity. However, the manuscript still states that the local observation is “Markovian for the decentralized policy” in the sense that the locally observable transition dynamics depend only on the current observation.

This statement is difficult to justify in a partially observable multi-agent environment. If agents outside the local observation influence future local states, two different hidden global states may generate the same

current observation but different future observation distributions. The local observation therefore need not constitute a Markov state.

I suggest simply stating that agents act on partial local observations and that the learning problem is solved approximately under this partial observability, without asserting a Markov property for the observation vector itself.

Relatedly, the terminology between a partially observable stochastic game and a Dec-POMDP should be made fully consistent. Parameter sharing and identical action spaces alone do not necessarily establish the common-reward structure normally associated with a Dec-POMDP. If individual agents receive distinct welfare rewards, the authors should either justify the Dec-POMDP formulation more explicitly or retain the more general partially observable stochastic-game terminology.

4. Reward shaping terminology should be made more precise.

The revised reward definition is now sufficiently clear, and this addresses my previous concern. However, the discussion connecting the reward to potential-based reward shaping should be qualified.

Formal policy-invariant potential-based shaping uses a term of the form

$$F(s, s') = \gamma\Phi(s') - \Phi(s),$$

whereas the manuscript uses a welfare difference of the form

$$\Phi(s') - \Phi(s).$$

The fact that $\gamma = 0.99$ is numerically close to one does not imply that the formal policy-invariance result continues to hold. I suggest describing the reward as being *motivated by* potential-difference shaping rather than suggesting that the policy-invariance property is approximately inherited.

5. The price analysis is improved, but “equilibrium” remains somewhat overstated.

The revision now makes an important and appropriate distinction: the reported $\sigma(t)$ measures cross-replicate consistency of cumulative geometric-mean price trajectories rather than instantaneous price dispersion within a market. This substantially improves the interpretation of the results.

However, this statistic alone does not constitute a formal test of convergence to an economic equilibrium. Indeed, the manuscript itself acknowledges that within-period price dispersion and a formal equilibrium benchmark remain outside the present analysis.

I therefore recommend restricting the language throughout to statements such as “more consistent cross-replicate price trajectories” or “more reproducible price formation.” Terms such as “equilibrium stability,” “equilibrium convergence,” or similar stronger claims should either be formally demonstrated or softened.

6. Several numerical and reporting inconsistencies should be corrected.

I noticed several internal inconsistencies that should be resolved before publication.

First, the top-10% wealth-share values reported in the text do not agree with those reported in Table 3. These values should be checked against the underlying data and made consistent.

Second, the manuscript refers in different places to approximately

$$5 \times 10^5$$

and

$$5 \times 10^6$$

training steps. The correct training horizon should be stated consistently throughout.

Third, with $N = 500$ agents and $R = 50$ replicates, the nominal initial cohort would contain 25,000 agents per condition. The pooled sample sizes reported in the Kaplan–Meier table are somewhat below this number and also differ across conditions. The reason for this discrepancy should be explained. If agents dying before the first recorded observation are absent from the reconstructed lifetime data, this should be explicitly addressed.

8. The interpretation of H4 should be revised.

The original hypothesis predicts lower inequality under trade-aware movement. The revised results do not appear to support this hypothesis as stated: terminal Gini coefficients are essentially unchanged, while under stochastic landscapes Theil's T is higher under DRL.

I would therefore prefer the manuscript to state that H4 is *not supported as originally formulated*, followed by the interesting exploratory finding that the distributional effects are more nuanced: the upper tail is compressed under deterministic landscapes, whereas stochastic environments exhibit an efficiency–equality trade-off.

This interpretation is both more accurate and, in my view, scientifically more interesting than describing H4 as “partially supported.”

Overall, I consider the revision a substantial improvement. The authors have responded seriously to the previous reviews, and several of the original methodological and presentation problems have been addressed convincingly. The remaining issues are nevertheless important, particularly the identification of the MRS/trade-awareness effect and the apparent mismatch in feasible movement sets.

If these points are resolved satisfactorily, I would be open to recommending publication in a subsequent round. Could you also use red in showing your revisions in the next round as this was highly difficult for me to see that changes took place.